

AI / ML Task 2

Feature Engineering, Model Optimization & Performance Comparison

Objective

The goal of this project was to improve the California Housing Price Prediction model by applying feature scaling and comparing multiple regression algorithms to identify the best-performing approach.

Dataset

- California Housing Dataset (scikit-learn)
- 20,640 samples
- 8 numerical features: MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude
- Target: Median House Value (HousePrice), in units of \$100,000

Models Evaluated

- Linear Regression
- Ridge Regression (alpha = 1.0)
- Decision Tree Regressor (max_depth = 5)

Preprocessing

Feature scaling was performed using **StandardScaler** to normalize feature values to zero mean and unit variance, improving stability and comparability across models. Data was split 80/20 into training and test sets with a fixed random seed (random_state = 42) for reproducibility.

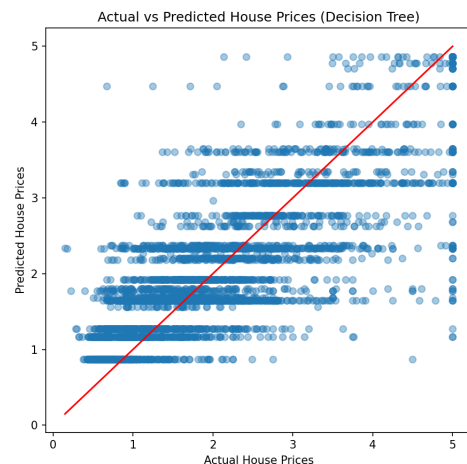
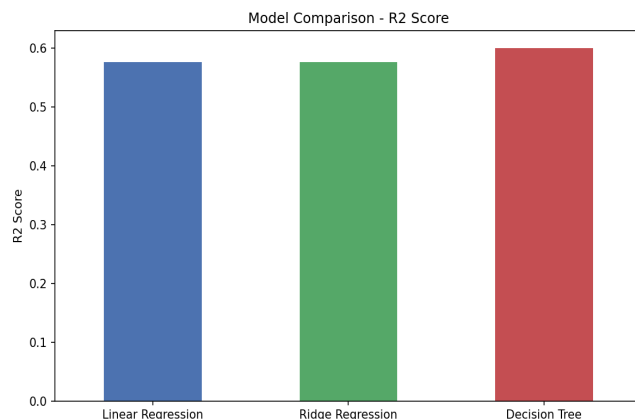
Evaluation Metrics

- **RMSE** (Root Mean Squared Error) — average prediction error in the target's units; lower is better
- **R2 Score** — proportion of variance in house prices explained by the model; closer to 1 is better

Results

Model	RMSE	R2 Score
Linear Regression	0.7456	0.5758
Ridge Regression	0.7456	0.5758
Decision Tree	0.7242	0.5997

Highlighted row indicates the best-performing model (lowest RMSE, highest R2).



Best Model

The **Decision Tree Regressor** (`max_depth = 5`) achieved the lowest RMSE (0.7242) and highest R2 score (0.5997) on the test set, making it the selected model for this task. It was retrained on the full training set and saved to **best_model.pkl** using `joblib`, alongside the fitted **scaler.pkl** needed to preprocess new inputs consistently.

Conclusion

Feature scaling and multi-model comparison provided a much clearer picture of model performance than evaluating a single algorithm alone. Ridge Regression performed virtually identically to plain Linear Regression here (RMSE 0.7456 for both), indicating that L2 regularization at $\alpha = 1.0$ had minimal effect — the underlying features are not highly collinear enough for regularization to make a meaningful difference. The depth-limited Decision Tree slightly outperformed both linear models, suggesting it captured some genuine non-linear structure in the data (e.g. interactions between income, location, and house age) without overfitting at `max_depth = 5`. Overall, the project demonstrates a professional machine learning workflow involving preprocessing, training, evaluation, comparison, and principled model selection.